4.2 Exponential Smoothing: Aggregate-Daily

2026-04-08

Goal

Apply exponential smoothing to the aggregate-daily Rossmann sales series and evaluate its forecasting performance using a holdout set.

We compare the ETS model to the observed holdout period and report holdout RMSE.

```r
library(tidyverse)
library(forecast)

daily_data <- readr::read_csv(
  "../../03_wrangling/daily_all_clean.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(
    Date = as.Date(Date)
  ) |>
  dplyr::arrange(Date)

head(daily_data)
```

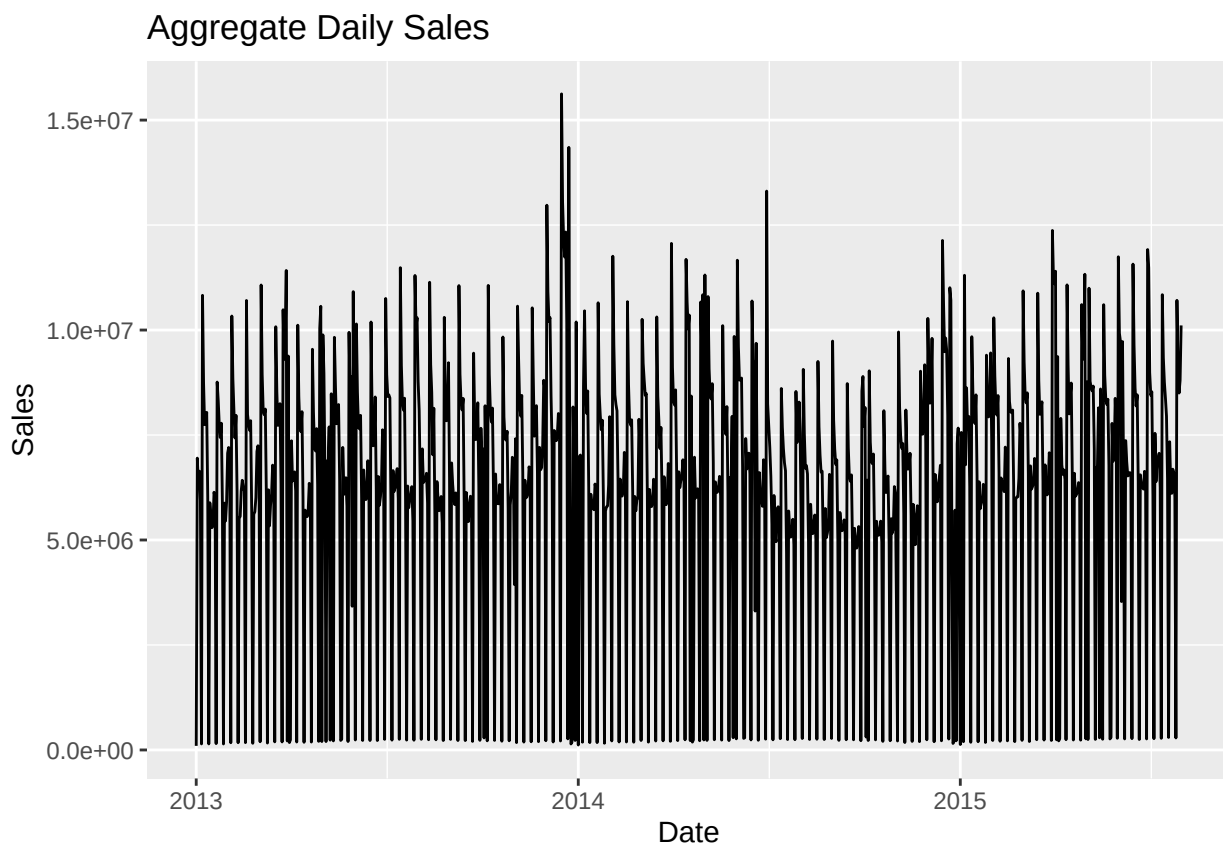
```
## # A tibble: 6 x 8
##   Date         Sales Customers PromoShare SchoolHolidayShare OpenStoreCount
##   <date>       <dbl>     <dbl>      <dbl>              <dbl>          <dbl>
## 1 2013-01-01   97235     19491          0              1                 17
## 2 2013-01-02 6949829    834075          0              1               1111
## 3 2013-01-03 6347820    761040          0              0.933           1109
## 4 2013-01-04 6638954    782137          0              0.933           1108
## 5 2013-01-05 5951593    687472          0              0.100           1107
## 6 2013-01-06  143904     26387          0              0.100             19
## # i 2 more variables: Weekday <chr>, IsWeekend <lgl>
```

```r
holdout_horizon <- 42
n_total <- nrow(daily_data)
n_train <- n_total - holdout_horizon

train_subset <- daily_data |> dplyr::slice(1:n_train)
holdout_subset <- daily_data |> dplyr::slice((n_train + 1):n_total)

holdout_split <- tibble::tibble(
  n_total = n_total,
  n_train = n_train,
  n_holdout = n_total - n_train,
  train_start = min(train_subset$Date),
  train_end = max(train_subset$Date),
  holdout_start = min(holdout_subset$Date),
  holdout_end = max(holdout_subset$Date)
```

```r
)

holdout_split
```

```
## # A tibble: 1 x 7
##   n_total n_train n_holdout train_start train_end  holdout_start holdout_end
##     <int>   <dbl>     <dbl> <date>      <date>     <date>        <date>
## 1     942     900        42 2013-01-01  2015-06-19 2015-06-20    2015-07-31
```

```r
ggplot(daily_data, aes(x = Date, y = Sales)) +
  geom_line() +
  labs(
    title = "Aggregate Daily Sales",
    x = "Date",
    y = "Sales"
  )
```




```r
sales_ts <- ts(train_subset$Sales, frequency = 7)
```

```r
ets_model <- ets(sales_ts)

ets_model
```

```
## ETS(M,N,M)
##
## Call:
## ets(y = sales_ts)
```

```
##
##   Smoothing parameters:
##     alpha = 0.1284
##     gamma = 1e-04
##
##   Initial states:
##     l = 5820592.519
##     s = 1.4163 0.0338 0.9776 1.1732 1.0903 1.1142
##            1.1946
##
##   sigma:  0.2468
##
##      AIC     AICc      BIC
## 31126.44 31126.68 31174.46
```

```
summary(ets_model)
```

```
## ETS(M,N,M)
##
## Call:
## ets(y = sales_ts)
##
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##            1.1946
##
##   sigma:  0.2468
##
##      AIC     AICc      BIC
## 31126.44 31126.68 31174.46
##
## Training set error measures:
##                     ME    RMSE     MAE       MPE     MAPE     MASE      ACF1
## Training set -36269.67 1899126 1186058 -101.6393 116.2889 0.567395 0.1705745
```

```
ets_holdout_fc <- forecast(ets_model, h = holdout_horizon)

holdout_predictions <- tibble::tibble(
  Date = holdout_subset$Date,
  actual = holdout_subset$Sales,
  predicted = as.numeric(ets_holdout_fc$mean)
)

head(holdout_predictions)
```

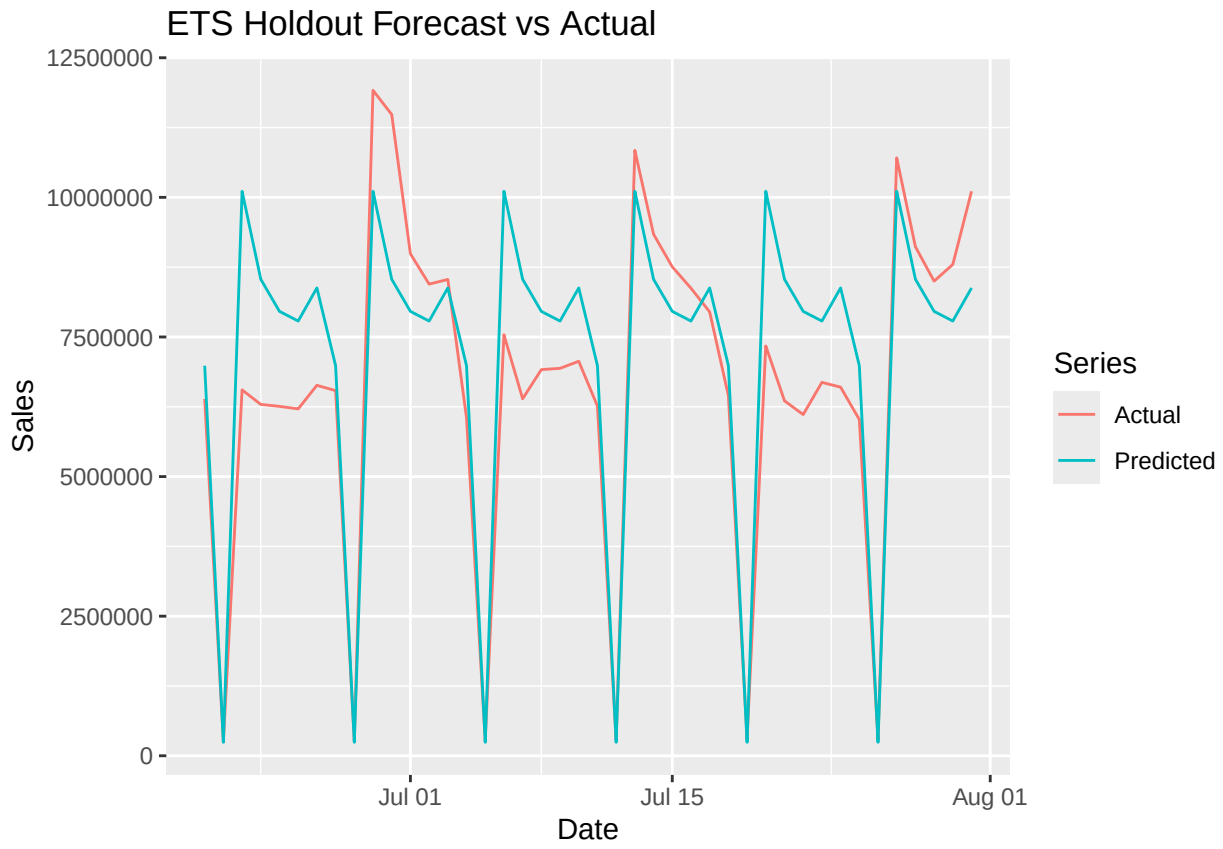
```
## # A tibble: 6 x 3
##   Date        actual predicted
##   <date>       <dbl>     <dbl>
## 1 2015-06-20 6390661  6985649.
## 2 2015-06-21  249849   241550.
```

```
## 3 2015-06-22 6552538 10109008.
## 4 2015-06-23 6292372  8533010.
## 5 2015-06-24 6257424  7958452.
## 6 2015-06-25 6211764  7785231.
```

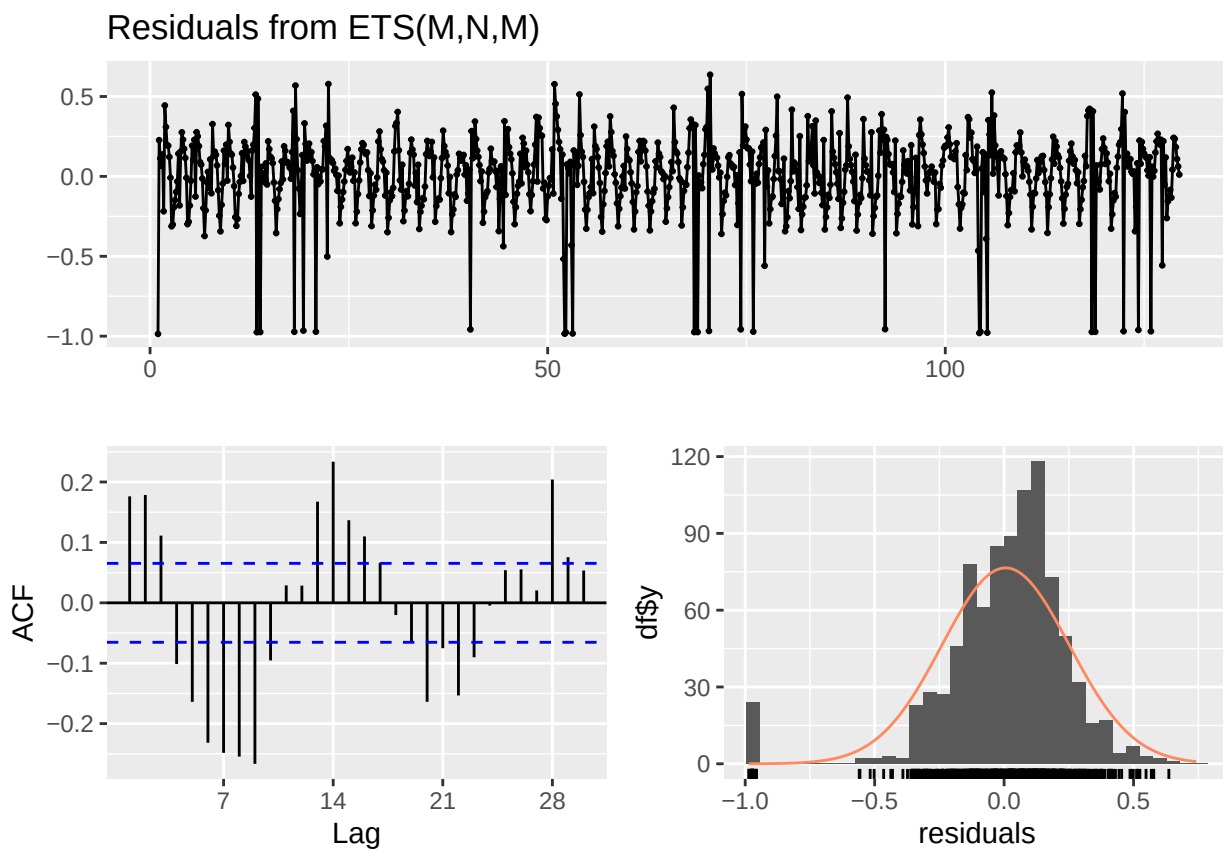
```r
rmse_vec <- function(actual, pred) {
  sqrt(mean((actual - pred)^2, na.rm = TRUE))
}

ets_holdout_rmse <- rmse_vec(
  actual = holdout_predictions$actual,
  pred = holdout_predictions$predicted
)

holdout_rmse_summary <- tibble::tibble(
  model = "ETS",
  holdout_rmse = ets_holdout_rmse,
  n_holdout = nrow(holdout_predictions)
)

holdout_rmse_summary
```

```
## # A tibble: 1 x 3
##   model holdout_rmse n_holdout
##   <chr>        <dbl>     <int>
## 1 ETS       1422687.        42
```

```r
ggplot(holdout_predictions, aes(x = Date)) +
  geom_line(aes(y = actual, color = "Actual")) +
  geom_line(aes(y = predicted, color = "Predicted")) +
  labs(
    title = "ETS Holdout Forecast vs Actual",
    x = "Date",
    y = "Sales",
    color = "Series"
  )
```




```
checkresiduals(ets_model)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(M,N,M)
## Q* = 415.64, df = 14, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 14
```

```r
dir.create("results", showWarnings = FALSE, recursive = TRUE)

readr::write_csv(holdout_split, "results/ets_holdout_split.csv")
readr::write_csv(holdout_predictions, "results/ets_holdout_predictions.csv")
readr::write_csv(holdout_rmse_summary, "results/ets_holdout_rmse_summary.csv")
```